# Multi-protocol Routing and Multi-service Network Configuration and Troubleshooting Lab or Configuring a Scalable Network with RIP, Static Routing, VLANs, DHCP, DNS, and Web Services.

## OBJECTIVES:

- **Core Routing**: Interconnect two Cisco 2911 routers via **Serial** WIC-2T links, implementing a hybrid of RIPv2 and Static routing for network reachability.
- **VLAN & L2 Setup**: Create four VLANs across 2960 switches and configure Router-on-a-Stick using sub-interfaces to enable Inter-VLAN routing.
- **DHCP & Relay**: Deploy a centralized DHCP server with four unique scopes, configuring ip helper-address on router sub-interfaces to relay requests across segments.
- **Services & Testing**: Host a DNS and Web server for domain name resolution and access, then verify connectivity using Cisco IOS show commands**.**

## THEORY:

1. **RIP version 2 (Routing Information Protocol)**: A distance-vector routing protocol that uses **hop count** as a metric. Version 2 supports **VLSM** (Variable Length Subnet Masking) and uses multicast for more efficient route updates.
2. **Static Routing**: A manual method of adding routes to a router's routing table. It is highly **secure** and consumes minimal CPU resources, as it does not rely on dynamic update algorithms.
3. **VLAN (Virtual Local Area Network)**: A logical segmentation of a physical network into multiple broadcast domains. This improves **security** and reduces network congestion by limiting broadcast traffic to specific groups.
4. **Router-on-a-Stick**: A configuration where a single physical router interface is divided into multiple **sub-interfaces** (logical interfaces). Each sub-interface handles traffic for a specific VLAN, enabling **Inter-VLAN routing**.
5. **DHCP (Dynamic Host Configuration Protocol)**: A network service that automatically assigns IP addresses, subnet masks, and default gateways to devices. A **DHCP Scope** is a specific range of addresses defined for a single subnet or VLAN.

6. **IP Helper-Address**: A command configured on a router interface to forward **UDP broadcast** traffic (like DHCP requests) to a server located on a different subnet or VLAN.
7. **DNS (Domain Name System)**: The "phonebook of the internet" that translates human-readable domain names (e.g., example.com) into numerical **IP addresses** using records like the **A record**.
8. **WIC-2T (WAN Interface Card)**: A hardware module for Cisco routers that provides two **serial ports**. It is used to create Point-to-Point WAN connections between geographically separated routers.
9. **802.1Q Encapsulation:** The IEEE standard for VLAN tagging on Ethernet frames. The encapsulation dot1Q command on a sub-interface tells the router which VLAN tag to expect and apply on that interface.
10. **Trunk Link:** A switch port that carries frames from multiple VLANs simultaneously using 802.1Q tags. The uplink from each switch Fa0/1 to its router is configured as a trunk.

## Network Topology & Device Inventory:

The network follows a hierarchical two-tier design. Two Cisco 2911 core routers are connected via a serial WAN link. Each router connects via a trunk link to a Cisco 2960 access switch hosting two VLANs. Servers and PCs connect to the switches as access port devices.

## Device Inventory

| Device | Model | Hostname | Role |
|---|---|---|---|
| Router 1 | Cisco 2911 | Router0 | Left-side gateway, RIP + Static routing |
| Router 2 | Cisco 2911 | Router1 | Right-side gateway, RIP + Static routing |
| Switch 1 | Cisco 2960-24TT | Switch0 | Left access switch, VLAN 10 & 30 |
| Switch 2 | Cisco 2960-24TT | Switch1 | Right access switch, VLAN 20 |

| Device | Model | Hostname | Role |
|---|---|---|---|
| Server 0 | Server-PT | Server0 | DHCP Server (multiple scopes) |
| Server 1 | Server-PT | Server1 | DNS + Web Server |
| PC 0 | PC-PT | PC0 | End device — VLAN 10 (left side) |
| PC 1 | PC-PT | PC1 | End device — VLAN 10 (left side) |
| PC 2 | PC-PT | PC2 | End device — VLAN 20 (right side) |
| PC 3 | PC-PT | PC3 | End device — VLAN 20 (right side) |

## Physical Cable Connections:

| From Device | Port | To Device | Port | Cable Type |
|---|---|---|---|---|
| Router0 | Gi0/0 | Router1 | Gi0/0 | Straight-through copper |
| Router0 | Gi0/1 | Switch0 | Fa0/1 | Straight-through copper |
| Router1 | Gi0/1 | Switch1 | Gi0/1 | Straight-through copper |
| Switch0 | Fa0/2 | PC0 | Fa0 | Straight-through copper |
| Switch0 | Fa0/3 | PC1 | Fa0 | Straight-through copper |
| Switch0 | Fa0/4 | Server0 | Fa0 | Straight-through copper |

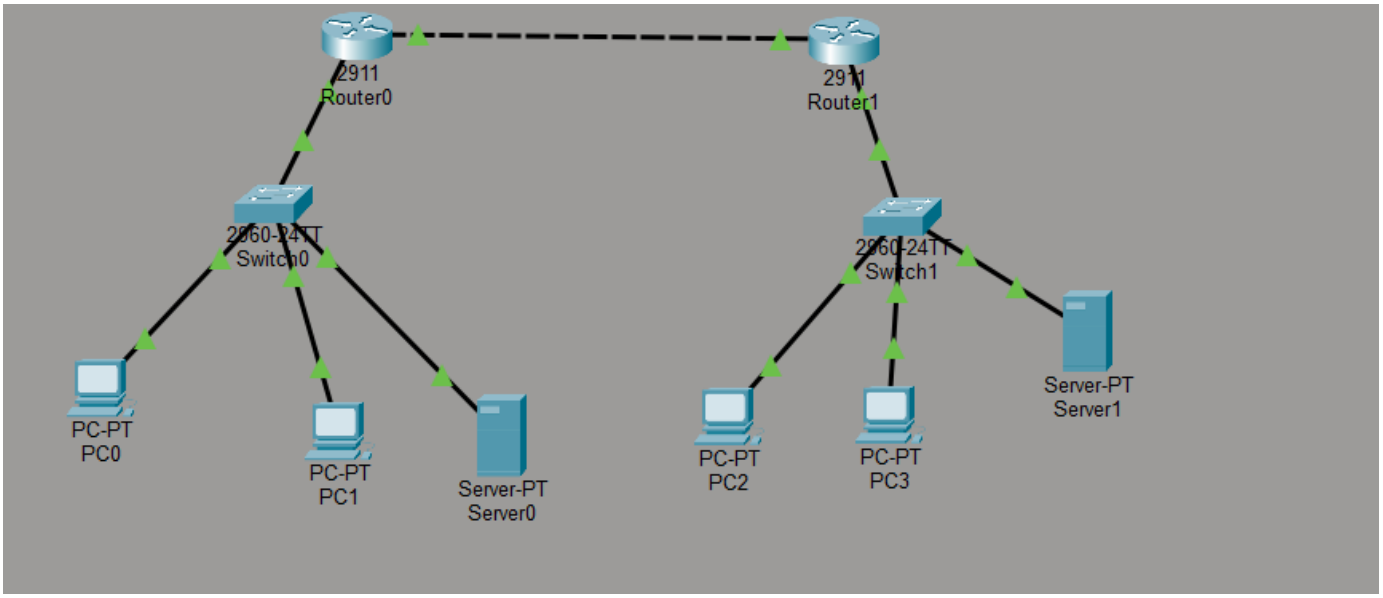
| From Device | Port | To Device | Port | Cable Type |
|---|---|---|---|---|
| Switch1 | Fa0/1 | PC2 | Fa0 | Straight-through copper |
| Switch1 | Fa0/2 | PC3 | Fa0 | Straight-through copper |
| Switch1 | Fa0/4 | Server1 | Fa0 | Straight-through copper |

## IP Addressing Table

| Device | Interface | IP Address | Subnet Mask | Purpose |
|---|---|---|---|---|
| Router0 | Gi0/0 | 10.0.12.1 | 255.255.255.252 | WAN link to Router1 |
| Router0 | Gi0/1 | 192.168.10.1 | 255.255.255.0 | Left LAN gateway |
| Router0 | Gi0/1.10 | 192.168.10.1 | 255.255.255.0 | VLAN 10 subinterface |
| Router0 | Gi0/1.20 | 192.168.20.1 | 255.255.255.0 | VLAN 20 subinterface |
| Router0 | Gi0/1.30 | 192.168.30.1 | 255.255.255.0 | VLAN 30 subinterface |
| Router1 | Gi0/0 | 10.0.12.2 | 255.255.255.252 | WAN link to Router0 |
| Router1 | Gi0/1 | 192.168.20.1 | 255.255.255.0 | Right LAN gateway |
| Server0 | Fa0 | 192.168.10.2 | 255.255.255.0 | DHCP Server |
| Server1 | Fa0 | 192.168.20.2 | 255.255.255.0 | DNS + Web Server |

| Device | Interface | IP Address | Subnet Mask | Purpose |
|---|---|---|---|---|
| PC0 | Fa0 | 192.168.10.10 | 255.255.255.0 | DHCP assigned |
| PC1 | Fa0 | 192.168.10.11 | 255.255.255.0 | DHCP assigned |
| PC2 | Fa0 | 192.168.20.10 | 255.255.255.0 | Static/DHCP |
| PC3 | Fa0 | 192.168.20.11 | 255.255.255.0 | Static/DHCP |

## Network Topology:




## Router Configuration:

### Router0(R0) Full Configuration

R0 is the DCE end of the serial link and the default gateway for VLAN 10 and VLAN 20. It runs RIP v2 and has a manual static route to VLAN 40 on the right side of the network.

**Interface Configuration:**

```
interface gi0/0
  ip address 10.0.12.1 255.255.255.252
  no shutdown
interface gi0/1
  no ip address
  no shutdown
interface gi0/1.10
  encapsulation dot1q 10
  ip address 192.168.10.1 255.255.255.0
  ip helper-address 192.168.10.2
interface gi0/1.20
  encapsulation dot1q 20
  ip address 192.168.20.1 255.255.255.0
  ip helper-address 192.168.10.2
interface gi0/1.30
  encapsulation dot1q 30
  ip address 192.168.30.1 255.255.255.0
  ip helper-address 192.168.10.2
```

**Static Route**

```
ip route 192.168.20.0 255.255.255.0 10.0.12.2
```

**RIP Configuration**

```
router rip
  version 2
  no auto-summary
  network 10.0.12.0
  network 192.168.10.0
```

## Router1(R1) Full Configuration:

R1 is the DTE end of the serial link and the gateway for VLAN 30 and VLAN 40. It runs RIP v2 and has a static route back to VLAN 10 where the DHCP and DNS server resides.

**Interface Configuration**

```
interface gi0/0
  ip address 10.0.12.2 255.255.255.252
  no shutdown
interface gi0/1
  ip address 192.168.20.1 255.255.255.0
  no shutdown
```

**Static Route**

```
ip route 192.168.10.0 255.255.255.0 10.0.12.1
```

**RIP Configuration**

```
router rip
  version 2
  no auto-summary
  network 10.0.12.0
  network 192.168.20.0
```

## Routing Protocols

### RIP Version 2

RIP (Routing Information Protocol) Version 2 was configured on both routers to enable dynamic route exchange. RIP v2 supports classless routing and sends updates every 30 seconds.

| Parameter | Router0 Value | Router1 Value |
|---|---|---|
| Protocol | RIP v2 | RIP v2 |
| Auto-summary | Disabled | Disabled |
| Update interval | 30 seconds | 30 seconds |
| Networks advertised | 10.0.12.0, 192.168.10.0 | 10.0.12.0, 192.168.20.0 |
| Route learned via RIP | 192.168.20.0/24 | 192.168.10.0/24 |
| Administrative distance | 120 | 120 |

### Static Routing

Static routes were configured on both routers as a backup and to demonstrate combined static + dynamic routing. The static route has an administrative distance of 1, lower than RIP (120), so it takes precedence.

| Router | Static Route | Next Hop | Purpose |
|---|---|---|---|
| Router0 | 192.168.20.0/24 | 10.0.12.2 | Reach right-side LAN via Router1 |
| Router1 | 192.168.10.0/24 | 10.0.12.1 | Reach left-side LAN via Router0 |

## RIP Verification Output

The following output was confirmed on both routers using show commands:

**Router0 — show ip rip database**

```
10.0.12.0/30     directly connected, GigabitEthernet0/0
192.168.10.0/24  directly connected, GigabitEthernet0/1
192.168.20.0/24  [1] via 10.0.12.2, 00:00:18, GigabitEthernet0/0
```

**Router1 — show ip rip database**

```
10.0.12.0/30     directly connected, GigabitEthernet0/0
192.168.20.0/24  directly connected, GigabitEthernet0/1
192.168.10.0/24  [1] via 10.0.12.1, 00:00:25, GigabitEthernet0/0
```

## VLAN Configuration

VLANs were created on both switches to segment network traffic. The 2960 switches are Layer 2 only, so all routing between VLANs is handled by the routers using subinterfaces (router-on-a-stick).

## VLAN Assignment Table

| VLAN ID | Name | Network | Switch | Ports | Devices |
|---|---|---|---|---|---|
| 10 | Left_LAN | 192.168.10.0/24 | Switch0 | Fa0/2, Fa0/3, Fa0/4 | PC0, PC1, Server0 |
| 20 | Right_LAN | 192.168.20.0/24 | Switch1 | Fa0/1, Fa0/2, Fa0/4 | PC2, PC3, Server1 |
| 30 | Management | 192.168.30.0/24 | Switch0 | Fa0/4 | Management devices |

## Switch0 VLAN Configuration

```
vlan 10
  name Left_LAN
vlan 30
  name Management
interface fa0/1
  switchport mode trunk
  switchport trunk allowed vlan 10,20,30
interface fa0/2
  switchport mode access
  switchport access vlan 10
interface fa0/3
  switchport mode access
  switchport access vlan 10
interface fa0/4
  switchport mode access
  switchport access vlan 10
```

## Switch1 VLAN Configuration

```
vlan 20
  name Right_LAN
vlan 30
  name Management
interface fa0/1
  switchport mode access
  switchport access vlan 20
interface fa0/2
  switchport mode access
  switchport access vlan 20
interface fa0/4
  switchport mode access
  switchport access vlan 20
```

## Inter-VLAN Routing

Since the 2960 switches are Layer 2 only, inter-VLAN routing was implemented using the router-on-a-stick method. Router0's Gi0/1 interface was divided into subinterfaces, each handling a separate VLAN using 802.1Q encapsulation.

### How Router-on-a-Stick Works

A single physical interface (Gi0/1) on Router0 carries traffic for all three VLANs. The switch port connected to the router is configured as a trunk port, allowing tagged VLAN traffic to pass. The router creates logical subinterfaces, one per VLAN, each with its own IP address serving as the default gateway for that VLAN.

| Subinterface | VLAN Tag | IP Address | Acts as Gateway For |
|---|---|---|---|
| Gi0/1.10 | 802.1Q VLAN 10 | 192.168.10.1 | All devices in VLAN 10 (PC0, PC1, Server0) |
| Gi0/1.20 | 802.1Q VLAN 20 | 192.168.20.1 | All devices in VLAN 20 (PC2, PC3, Server1) |
| Gi0/1.30 | 802.1Q VLAN 30 | 192.168.30.1 | Management VLAN devices |

## Verification

Router0 — show ip interface brief (confirmed output)

```
GigabitEthernet0/0     10.0.12.1    YES manual up    up
GigabitEthernet0/1     unassigned   YES manual up    up
GigabitEthernet0/1.10  192.168.10.1 YES manual up    up
GigabitEthernet0/1.20  192.168.20.1 YES manual up    up
GigabitEthernet0/1.30  192.168.30.1 YES manual up    up
```

## DHCP Server — Multiple Scopes

Server0 was configured as the DHCP server for the network. Three separate DHCP scopes were created — one for each VLAN. The ip helper-address command on Router0's subinterfaces forwards DHCP broadcast requests from clients to the DHCP server.

## Server0 Static IP Configuration

| Field | Value |
|---|---|
| IP Address | 192.168.10.2 |
| Subnet Mask | 255.255.255.0 |
| Default Gateway | 192.168.10.1 |
| DNS Server | 192.168.20.2 |

## DHCP Scope Definitions

| Pool Name | Network | Gateway | DNS Server | Start IP | Max Users |
|---|---|---|---|---|---|
| VLAN10_Pool | 192.168.10.0/24 | 192.168.10.1 | 192.168.20.2 | 192.168.10.10 | 50 |
| VLAN20_Pool | 192.168.20.0/24 | 192.168.20.1 | 192.168.20.2 | 192.168.20.10 | 50 |
| VLAN30_Pool | 192.168.30.0/24 | 192.168.30.1 | 192.168.20.2 | 192.168.30.10 | 50 |

## DHCP Helper Address Configuration

The ip helper-address command was applied on each Router0 subinterface to relay DHCP requests from clients to Server0:

```
interface gi0/1.10
  ip helper-address 192.168.10.2
interface gi0/1.20
  ip helper-address 192.168.10.2
interface gi0/1.30
  ip helper-address 192.168.10.2
```

## DNS and Web Server Configuration

### Server1 Static IP Configuration

| Field | Value |
|---|---|
| IP Address | 192.168.20.2 |
| Subnet Mask | 255.255.255.0 |
| Default Gateway | 192.168.20.1 |
| DNS Server | 192.168.20.2 (self) |

## DNS Service Configuration

The DNS service was enabled on Server1 with the following A record to resolve the lab domain name to Server1's IP address.

| Record Type | Name | IP Address | Purpose |
|---|---|---|---|
| A Record | www.lab.local | 192.168.20.2 | Resolves web domain to Server1 |

## Web (HTTP) Service

The HTTP service on Server1 was confirmed active. The default Cisco Packet Tracer index.html page was served and accessible from PC0 using the web browser at http://www.lab.local.

## Troubleshooting and Issues Resolved

### Issues Encountered and Fixes

| Issue | Symptom | Root Cause | Fix Applied |
|---|---|---|---|
| Server1 unreachable | ping 192.168.20.2 = 0/5 from Router1 | Server1 cable on Fa0/4 was in VLAN 1 (default), not VLAN 20 | Ran switchport access vlan 20 on Switch1 Fa0/4 |
| DNS not resolving | www.lab.local host not found | PC DNS server field was blank (0.0.0.0) | Set DNS server to 192.168.20.2 on all PCs |
| Trunk not applying | show interfaces trunk showed blank | Trunk configured on wrong port (Gi0/1 but router was on Fa0/1) | Verified correct port via show mac address-table |
| 1 packet loss on first ping | Request timed out on first packet only | Normal ARP resolution delay on first contact | Not an error — expected behavior in Packet Tracer |
| www.lab.local slow ping | Long delay before DNS reply | DNS resolution + RIP convergence time | Waited for RIP to converge; confirmed routes with show ip rip database |

## Show Commands Used for Troubleshooting

### Router Commands

| Command | Purpose | What to Look For |
|---|---|---|
| show ip route | View routing table | C (connected), S (static), R (RIP) routes present |
| show ip protocols | Verify RIP configuration | Version 2, correct networks, neighbor 10.0.12.x |
| show ip rip database | Check RIP learned routes | Remote networks with [1] via next-hop IP |
| show ip interface brief | Check interface status | All configured interfaces show up/up |
| show interfaces gi0/0 | Detailed interface info | Line protocol is up, correct IP |
| show arp | Check ARP table | Remote device MACs resolved on correct interface |
| ping <ip> | Test connectivity | 5/5 success (first ping may lose 1 due to ARP) |

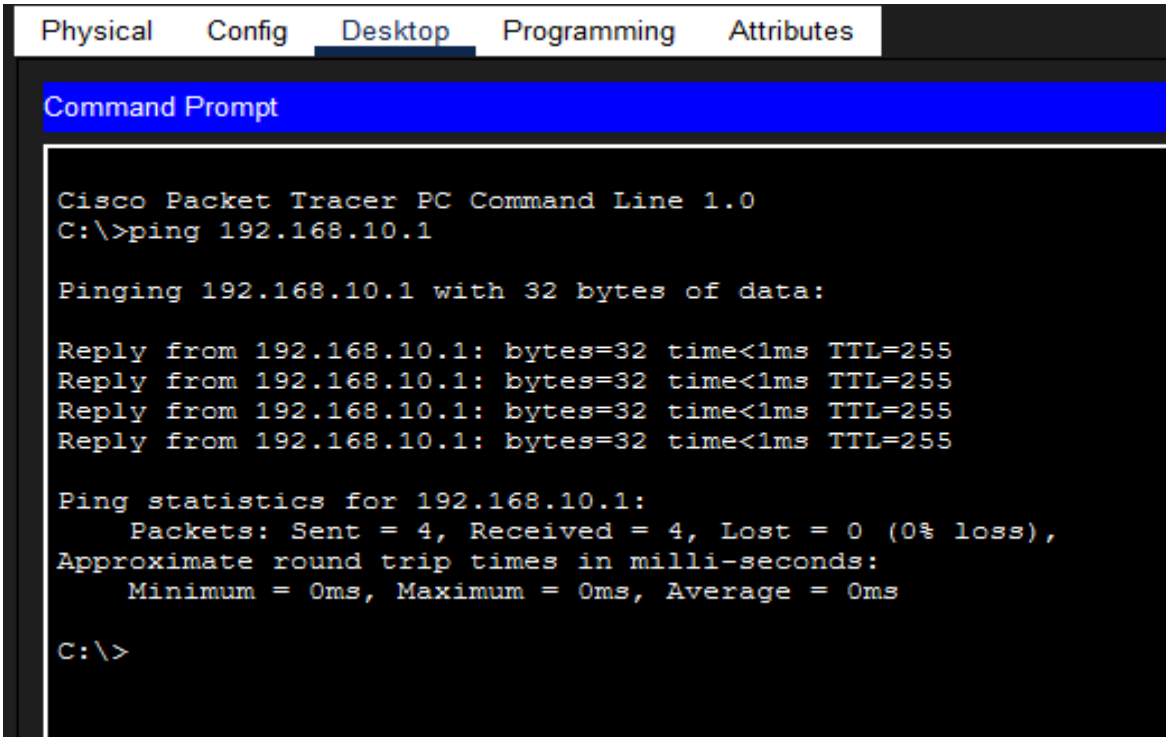
### Switch Commands

| Command | Purpose | What to Look For |
|---|---|---|
| show vlan brief | View VLAN database | All VLANs active with correct ports assigned |
| show interfaces trunk | Verify trunk ports | Uplink port listed with correct VLANs allowed |

| Command | Purpose | What to Look For |
|---|---|---|
| show mac address-table | Check MAC-to-port mappings | Devices appear on correct VLAN and port |
| show interfaces fa0/x switchport | Verify port mode | Access Mode VLAN: correct VLAN number |
| show interfaces fa0/x | Physical interface status | Interface is up, line protocol is up |

## Connectivity Test Results:

1. Gateway Reachability:




2. Router-to-Router Link:

```
Router0>ping 10.0.12.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.12.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0 ms

Router0>
```

3. End-to-End PC ping:

```
C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.20.10: bytes=32 time<1ms TTL=126
Reply from 192.168.20.10: bytes=32 time<1ms TTL=126
Reply from 192.168.20.10: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

4. Cross-Router ping:

```
C:\>ping 192.168.20.1

Pinging 192.168.20.1 with 32 bytes of data:

Reply from 192.168.20.1: bytes=32 time<1ms TTL=254
Reply from 192.168.20.1: bytes=32 time<1ms TTL=254
Reply from 192.168.20.1: bytes=32 time<1ms TTL=254
Reply from 192.168.20.1: bytes=32 time<1ms TTL=254

Ping statistics for 192.168.20.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

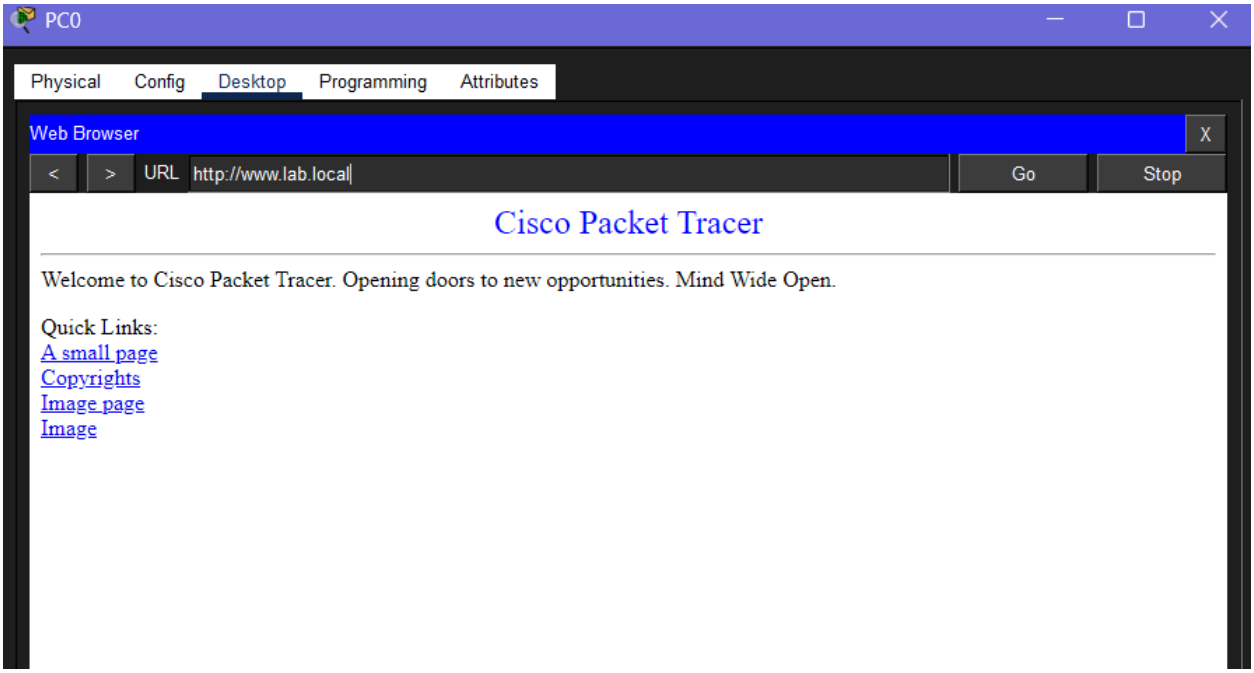
5. Server1 Reachability:

```
C:\>ping 192.168.20.2

Pinging 192.168.20.2 with 32 bytes of data:

Reply from 192.168.20.2: bytes=32 time=18ms TTL=126
Reply from 192.168.20.2: bytes=32 time<1ms TTL=126
Reply from 192.168.20.2: bytes=32 time<1ms TTL=126
Reply from 192.168.20.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 18ms, Average = 4ms

C:\>
```

6. DNS Resolution:

```
C:\>ping www.lab.local

Pinging 192.168.20.2 with 32 bytes of data:

Reply from 192.168.20.2: bytes=32 time=10ms TTL=126
Reply from 192.168.20.2: bytes=32 time=1ms TTL=126
Reply from 192.168.20.2: bytes=32 time=1ms TTL=126
Reply from 192.168.20.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 3ms

C:\>
```

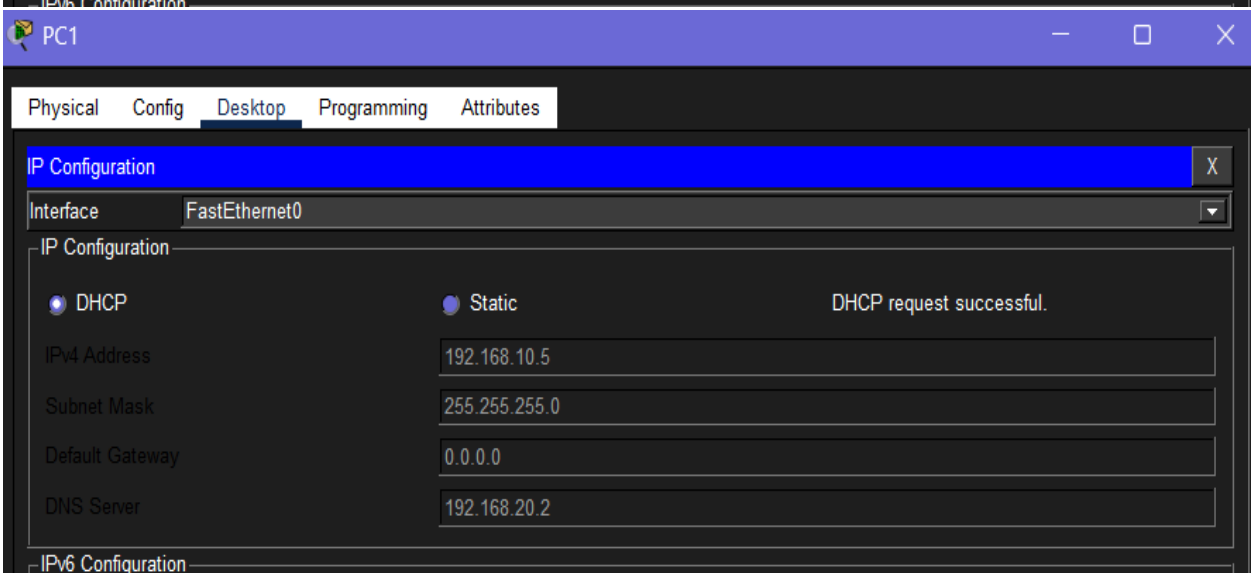
7. Web Server Access:




8. DHCP Helper:

```
Router0#enable
Router0#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router0(config)#interface gi0/1
Router0(config-if)#ip helper-address 192.168.10.2
Router0(config-if)#exit
Router0(config)#end
Router0#write memory
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
Router0#
```

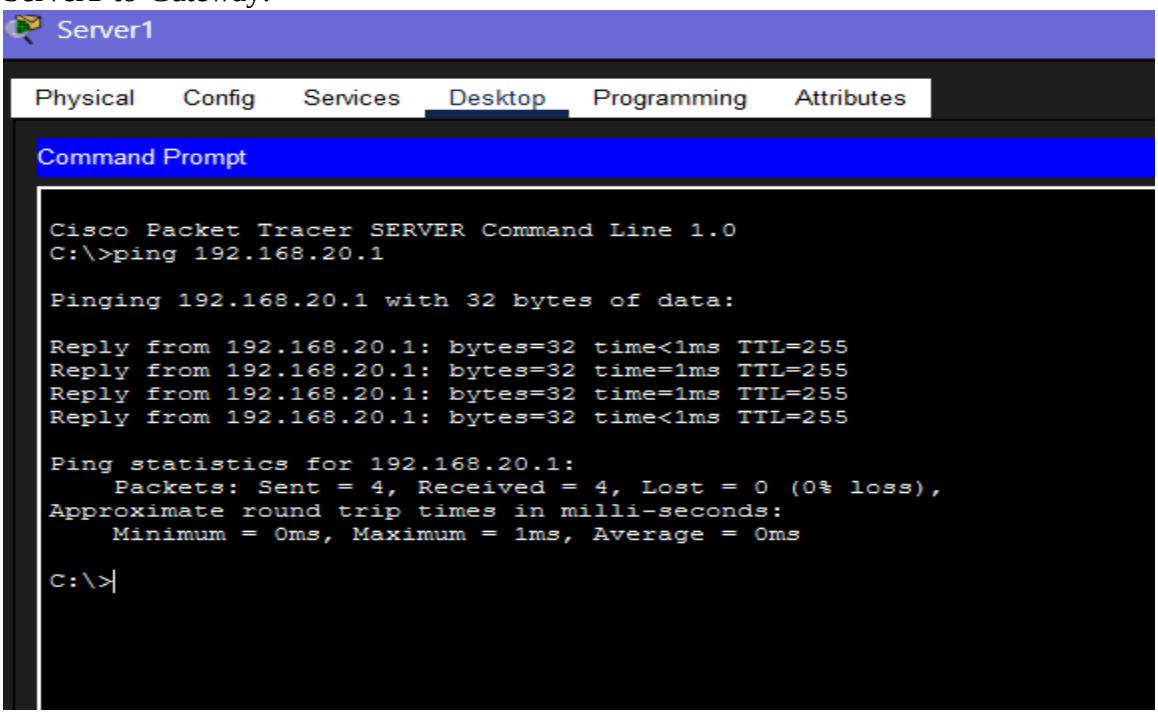
9. DHCP Assignment:




10. Router-to- Server1:

```
Router1#ping 192.168.20.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.20.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/1/8 ms

Router1#
```

11. Server1-to-Gateway:




## DISCUSSION AND CONCLUSION:

This lab successfully demonstrated the configuration of a dual-router network in Cisco Packet Tracer, integrating RIP v2 dynamic routing alongside static routes, VLAN segmentation across two 2960 switches, inter-VLAN routing via router-on-a-stick subinterfaces, a multi-scope DHCP server with helper addresses, and DNS and web services accessible across the network. Several real-world troubleshooting scenarios were encountered, including a misassigned VLAN port on Switch1 that prevented Server1 from communicating, which was resolved using show mac address-table and show vlan brief to pinpoint the fault. All connectivity tests ultimately passed, confirming that the network was fully operational end-to-end, and the lab reinforced the importance of methodical verification using Cisco show commands at each stage of configuration.